

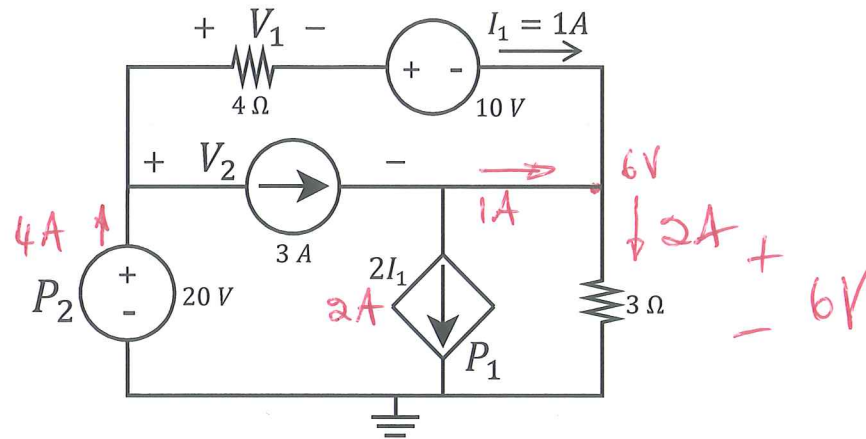
VRAAG/QUESTION 1 [18]

Vraag 1.1 [4]

Bepaal P_1 , P_2 , V_1 en V_2 .

Question 1.1 [4]

Determine P_1 , P_2 , V_1 and V_2 .



$$V_1 = 1 \times 4 = 4V \quad \checkmark$$

$$V_2 = 20 - 6 = 14V \quad \checkmark$$

$$P_1 = 2 \times 6 = 12W \quad \checkmark$$

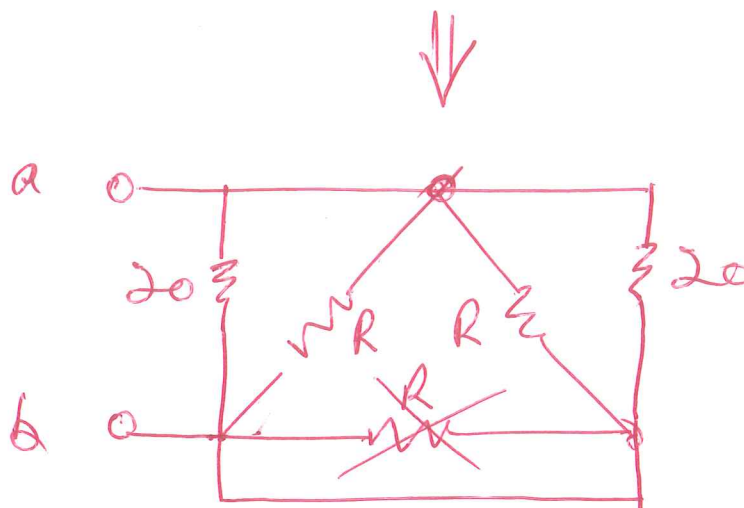
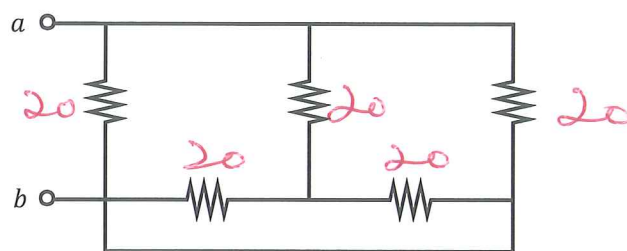
$$P_2 = -20 \times 4 = -80W \quad \checkmark$$

Vraag 1.2 [2]

Bepaal die ekwivalente weerstand van onderstaande kringbaan ten opsigte van terminale a en b .
Aanvaar alle weerstande het 'n waarde van $20\ \Omega$.

Question 1.2 [2]

Determine the equivalent resistance of the circuit below with respect to terminals a and b . Assume all resistors have a value of $20\ \Omega$.



$$R = \frac{20 \times 20 + 20 \times 20 + 20 \times 20}{20}$$

$$= 60\ \Omega$$

$$\therefore R_{ab} = 20 \parallel 60 \parallel 60 \parallel 20$$

$$= 7.5\ \Omega \quad \checkmark \checkmark$$

Vraag 1.3 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$aV_1 + bV_2 = 80$$

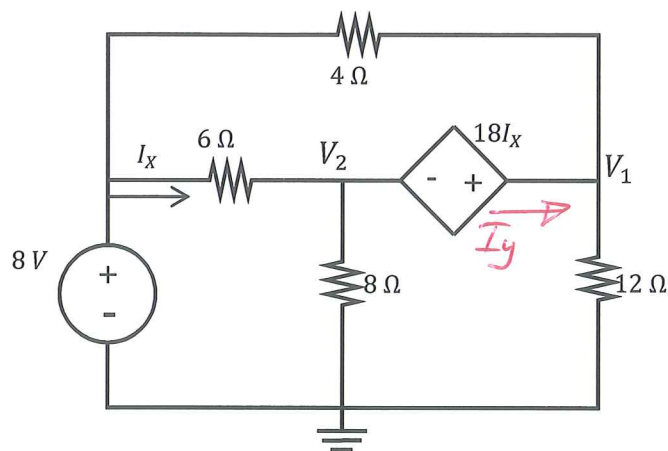
$$cV_1 + dV_2 = 24$$

Question 1.3 [4]

Use nodal analysis to set up two equations of the following form:

$$aV_1 + bV_2 = 80$$

$$cV_1 + dV_2 = 24$$



$$V_1 - V_2 = 18 I_x = 18 \left(\frac{8 - V_2}{6} \right) \\ = 3(8 - V_2)$$

$$\therefore V_1 + 2V_2 = 24$$

$$c = 1; d = 2$$

$$\frac{8 - V_2}{6} - \frac{V_2}{8} = I_y = \frac{V_1 - 8}{4} + \frac{V_1}{12}$$

$$\times 48: 8(8 - V_2) - 6V_2 = 12(V_1 - 8) + 4V_1$$

$$\therefore 16V_1 + 14V_2 = 160$$

$$8V_1 + 7V_2 = 80$$

$$a = 8; b = 7$$

Vraag 1.4 [4]

Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:

$$ei_2 + fi_3 = 60$$

$$gi_2 + hi_3 = 10$$

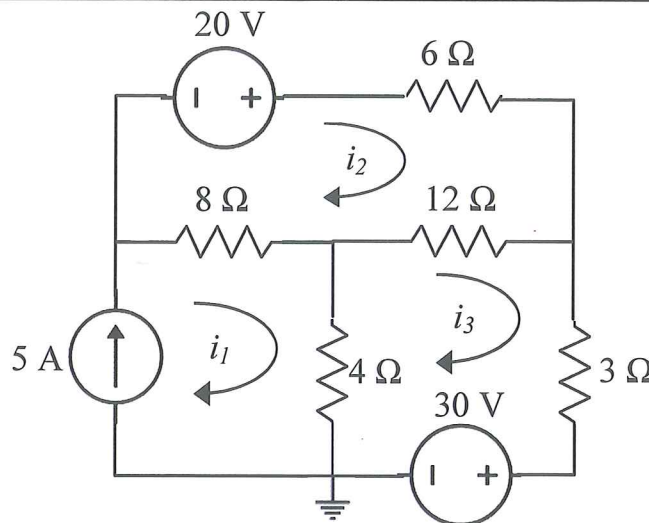
Question 1.4 [4]

Use mesh analysis to set up two equations of the following form:

$$ei_2 + fi_3 = 60$$

$$gi_2 + hi_3 = 10$$

$$i_1 = 5A$$



$$20 = 6i_2 + 12(i_2 - i_3) + 8(i_2 - 5)$$

$$\therefore 26i_2 - 12i_3 = 60$$

$$e = 26 \checkmark; \quad f = -12 \checkmark$$

$$30 = 12(i_2 - i_3) + 4(5 - i_3) + 3(-i_3)$$

$$30 - 20 = 12i_2 + i_3(-12 - 4 - 3)$$

$$\therefore 12i_2 - 19i_3 = 10$$

$$g = 12 \checkmark; \quad h = -19 \checkmark$$

Vraag 1.5 [4]Gegee: $\beta = 100$, $V_{BE} = 0.7 \text{ V}$.

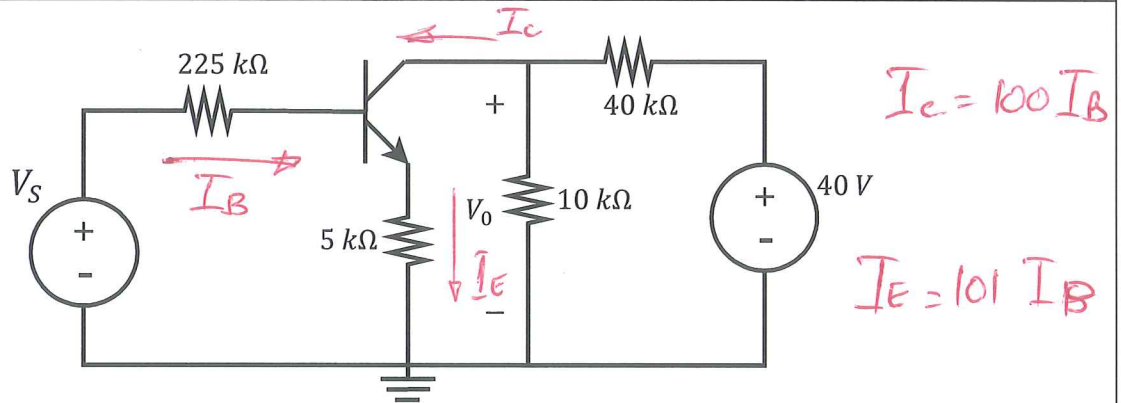
- a) Indien $V_S = 8 \text{ V}$, bepaal I_B .
 b) Indien $I_B = 5 \mu\text{A}$, bepaal V_0 .

(Vrae a) en b) is onafhanklik van mekaar.)

Question 1.5 [4]Given: $\beta = 100$, $V_{BE} = 0.7 \text{ V}$.

- a) If $V_S = 8 \text{ V}$, determine I_B .
 b) If $I_B = 5 \mu\text{A}$, determine V_0 .

(Questions a) and b) are independent of one another.)



$$\begin{aligned} \text{a) } V_S = 8 &= 225\text{k} \times I_B + 0.7 + 5\text{k} \times I_E \\ &= 225\text{k} \times I_B + 5\text{k} \times 101 \times I_B + 0.7 \end{aligned}$$

$$\therefore (225\text{k} + 505\text{k}) \times I_B = 8 - 0.7$$

$$730\text{k} \times I_B = 7.3$$

$$\therefore I_B = \frac{7.3}{730\text{k}} = 10\mu\text{A} \quad \checkmark\checkmark$$

$$\text{b) } I_B = 5\mu\text{A} \quad \therefore I_C = 0.5\text{mA}$$

$$\therefore 0.5\text{mA} + \frac{V_0}{10\text{k}} + \frac{V_0 - 40}{40\text{k}} = 0$$

$$\times 400\text{k} : 200 + 40V_0 + 10(V_0 - 40) = 0$$

$$\therefore 50V_0 = 200$$

$$V_0 = 4\text{V} \quad \checkmark\checkmark$$

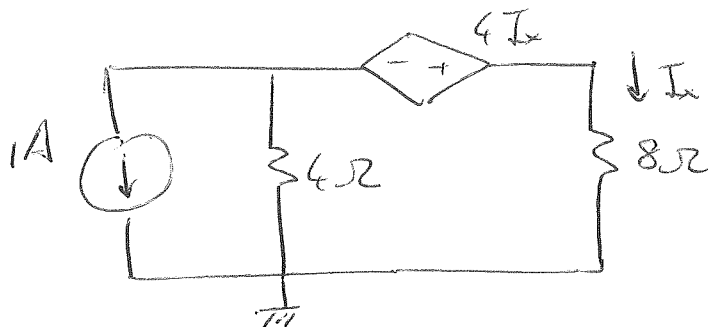
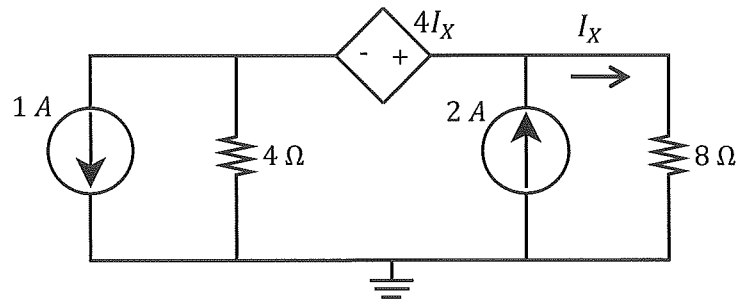
VRAAG/QUESTION 2 [12]

Question 2.1 [2]

Use superposition and determine the contribution to I_x by the 1 A current source only.

Vraag 2.1 [2]

Maak gebruik van superposisie en bepaal die bydrae tot I_x slegs as gevolg van die 1 A stroombron.

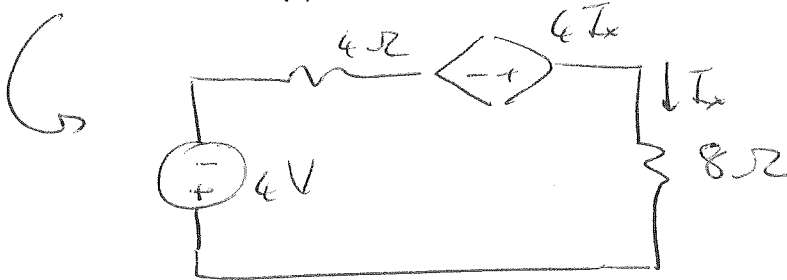


$$\rightarrow * 1 + \frac{V_1}{4} + \frac{V_1 + 4I_x}{8} = 0$$

$$* V_1 = 8I_x - 4I_x = 4I_x$$

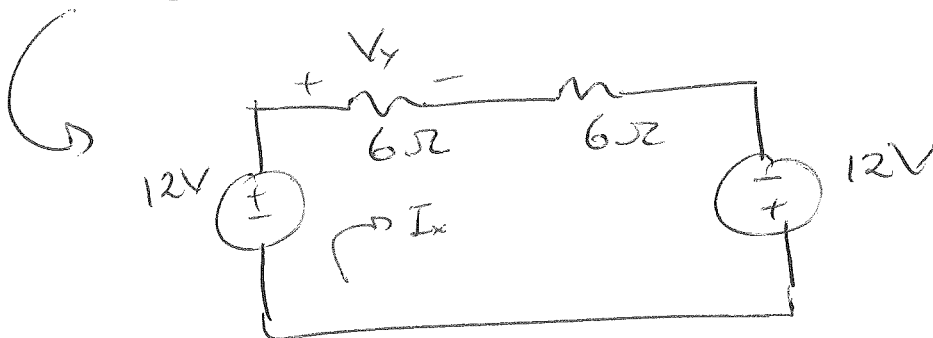
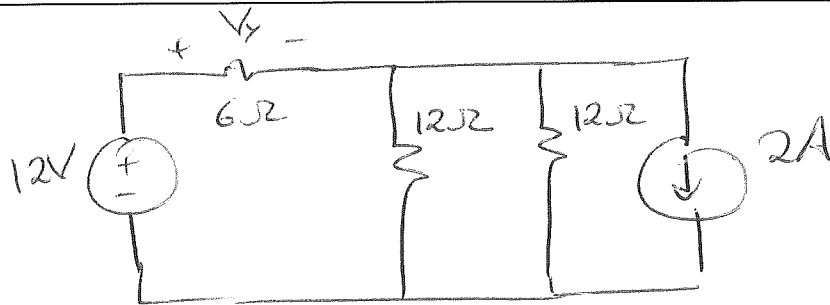
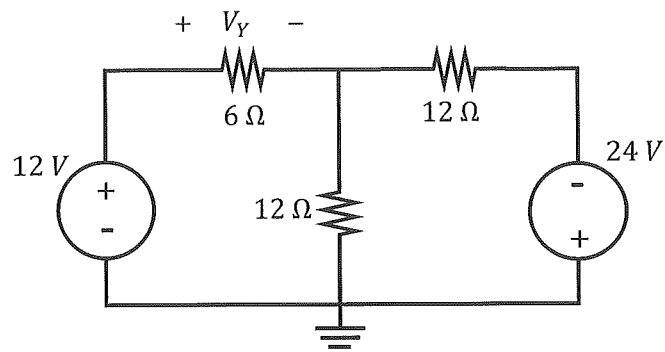
$$\therefore 1 + I_x + \frac{8I_x}{8} = 0$$

$$I_x = -0.5 A$$



$$+4 + 4I_x - 4I_x + 8I_x = 0$$

$$I_x = -0.5 A \quad \checkmark \checkmark$$

Question 2.2 [2]Make use of source transformation and determine V_Y .**Vraag 2.2 [2]**Maak gebruik van brontransformatie en bepaal V_Y .

$$\ast \quad -12 + 6 I_x + 6 I_x - 12 = 0$$

$$I_x = 2A$$

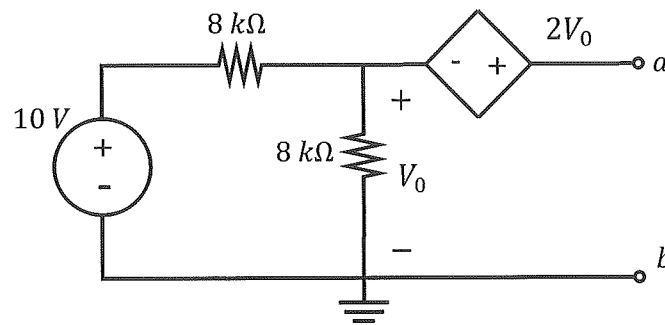
$$\ast \quad V_Y = 2 \times 6 = \underline{12V}$$

Question 2.3 [4]

Determine the Thevenin equivalent voltage V_{TH} and resistance R_{TH} wrt terminals $a-b$.

Vraag 2.3 [4]

Bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} tov terminale $a-b$.

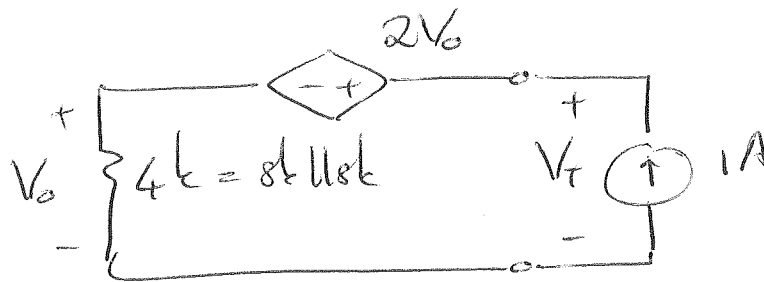


V_{TH}

$$V_0 = \frac{10 \times 8k}{8k + 8k} = 5V \quad (\rightarrow V\text{-division})$$

(KVL) $\therefore V_{ab} = 2V_0 + V_0 = \underline{15V}$ ✓

R_{TH}



KVL: $-V_T + 2V_0 + V_0 = 0$

5

$$V_T = 3V_0$$

$$= 3(4k \times 1)$$

$$= 12kV$$

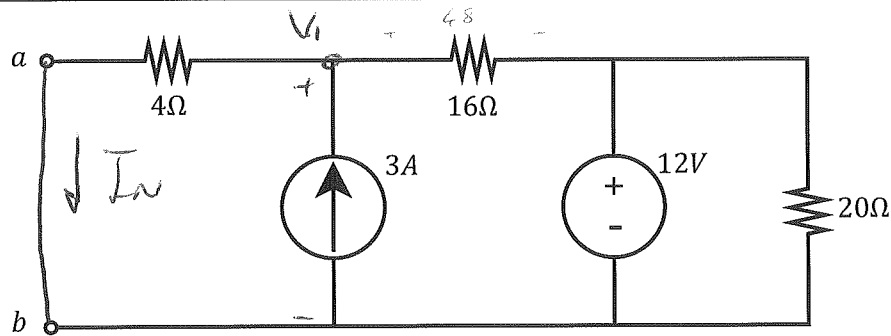
$$\therefore R_{TH} = \frac{12kV}{1A} = \underline{12k\Omega}$$
 ✓

Question 2.4 [4]

Determine the Norton equivalent current I_N and resistance R_N wrt terminals $a-b$.

Vraag 2.4 [4]

Bepaal die Norton ekwivalente stroom I_N en weerstand R_N tov terminale $a-b$.



$$\underline{I_N} \quad \frac{V_1}{4} + \frac{V_1 - 12}{16} = 3.$$

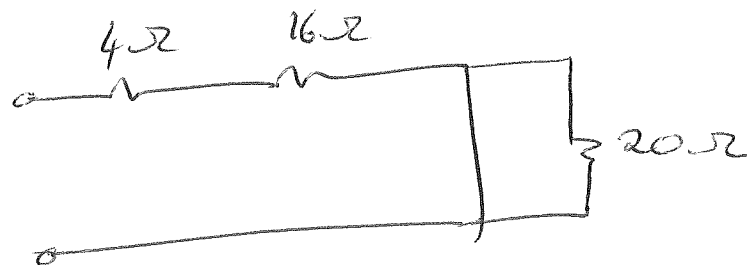
$$(\times 16) \quad 4V_1 + V_1 - 12 = 48$$

$$5V_1 = 60$$

$$V_1 = 12V$$

$$\therefore I_N = \frac{12}{4} = \underline{3A} \quad \checkmark$$

R_N



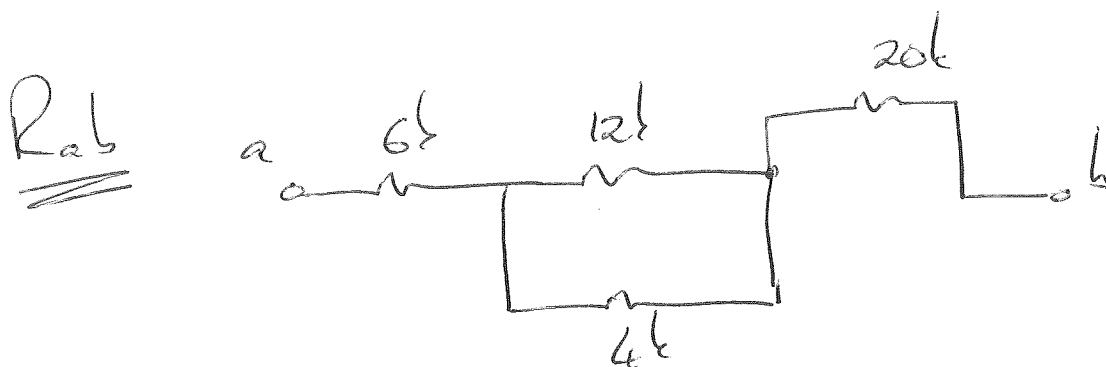
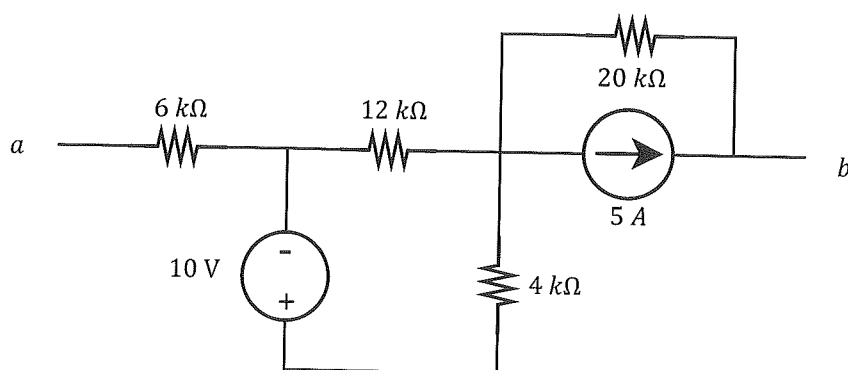
$$R_N = 4 + 16 = \underline{20\Omega} \quad \checkmark$$

Question 2.5 [2]

What loadresistance must be coupled between terminals a and b such that maximum power is delivered to the load?

Vraag 2.5 [2]

Watter lasweerstand moet tussen terminale a en b gekoppel word sodat maksimum drywing oorgedra word aan die las?



$$\begin{aligned}
 R_L = R_N = R_{ab} &= (6 + 12 \parallel 4 + 20) \text{ k} \\
 &= \left(26 + \frac{12 \times 4}{12 + 4} \right) \text{ k} \\
 &= (26 + 3) \text{ k} \\
 &= \underline{29 \text{ k}\Omega} \quad \checkmark
 \end{aligned}$$

EBN111 SEMESTER TEST/ TOETS 3 (16 MAY 2016)
ANSWER SHEET / ANTWOORDBLAD

Surname: Van	D. V.	Student number: Studentennummer:						
Initials: Voorletters:	BHATT	Group: Groep:	ENG	ENG (Mech)	AFR			

Q1		Q2		Q3		Tot:	/45
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QUESTION 1 - VRAAG 1 (18)				
1.1	$P_1 =$	$P_2 =$	$V_1 =$	$V_2 =$
1.2	$R_{ab} =$			
1.3	$a =$	$b =$	$c =$	$d =$
1.4	$e =$	$f =$	$g =$	$h =$
1.3	$a) I_B =$		$b) V_0 =$	

QUESTION 2 - VRAAG 2 (14)	
2.1	$I_{X(1A)} =$
2.2	$V_Y =$
2.3	$V_{TH} =$ $R_{TH} =$
2.4	$I_N =$ $R_N =$
2.5	$R_L =$

VRAAG 3 - QUESTION 3 (16)	
3.1	$i_f = 2 \mu A$
3.2	$a) v_{o1} = 2 mV$ $b) v_{o2} = 16 mV$ $c) v_o = 442 mV$
3.3	$a) w_C = 225 \mu J$ $b) i_L = 7.5 mA$
3.4	$L_{eq} = 0.061 H$ or $61 mH$
3.5	$v_L(t) = -1.44 e^{-0.2t} mV$
3.6	$Z_{rect} = 4.72 - j133 \Omega$

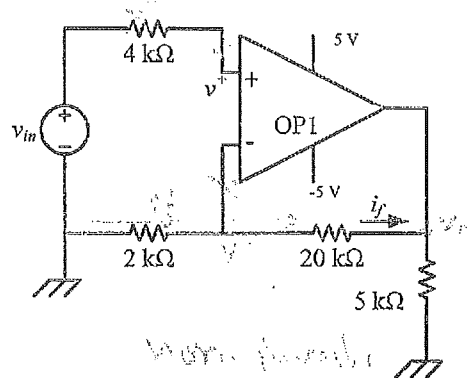
VRAAG/QUESTION 3 [16]

Question 3.1 [2]

Calculate the current i_f for $v^+ = -4\text{mV}$.

Vraag 3.1 [2]

Bereken die stroom i_f indien $v^+ = -4\text{mV}$.



$$i_f = \frac{0 - v^-}{2k}$$

$$v^- = v^+ = v_{in} = -4\text{mV}$$

$$i_f = \frac{-(-4\text{m})}{2k} = \underline{\underline{2\mu\text{A}}}$$

Question 3.2 [4]

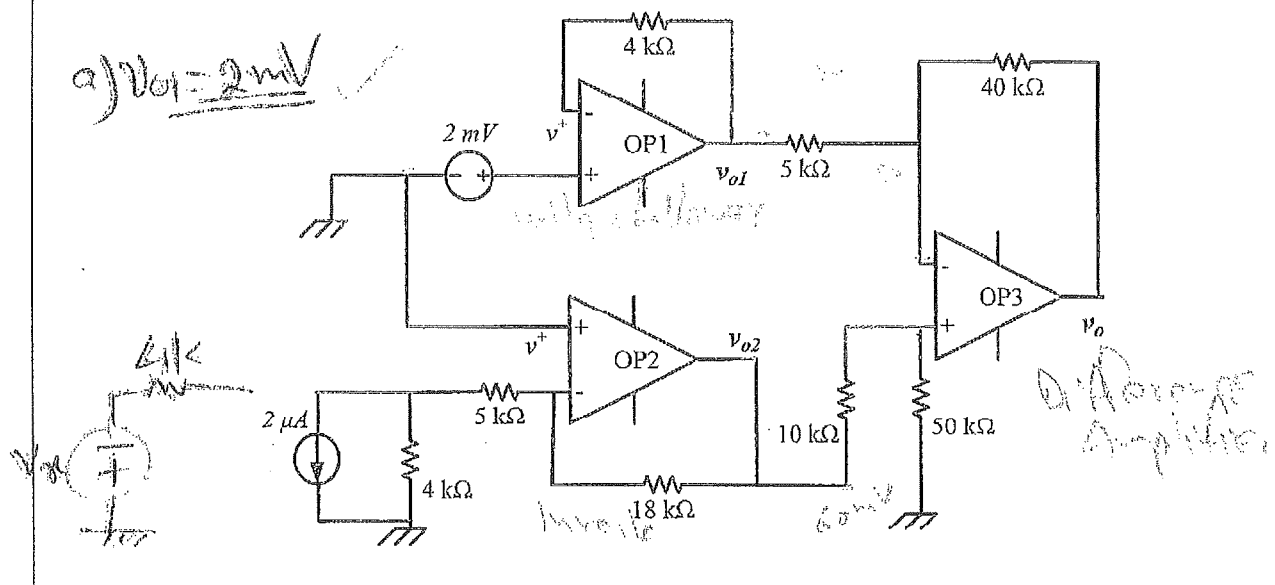
Given $V_{cc} = 5V$. Calculate:

- v_{o1}
- v_{o2}
- v_o when $v_{o1} = 1\text{ mV}$ and $v_{o2} = 60\text{ mV}$

Vraag 3.2 [4]

Gegee $V_{cc} = 5V$. Bereken:

- v_{o1}
- v_{o2}
- v_o wanneer $v_{o1} = 1\text{ mV}$ en $v_{o2} = 60\text{ mV}$



$$b) v_{o2} = \frac{-12k}{R_{in}} \times V_x$$

$$R_{in} = 4k + 5k = 9k\Omega$$

$$V_x = -2\mu A \times 4k = -8\text{ mV}$$

$$\therefore v_{o2} = \frac{-12k}{9k} \times -8\text{ mV} = \underline{16\text{ mV}}$$

c)

$$v_{op3}^+ = \frac{50k}{50k + 10k} \times 60\text{ mV} = 50\text{ mV}$$

KCL @ v_{op3}^+

$$\frac{1\text{ mV} - 50\text{ mV}}{5k} = \frac{50\text{ mV} - v_o}{40k}$$

$$-49\text{ mV} \times 8 = 50\text{ mV} - v_o$$

$$v_o = 50\text{ mV} + 392\text{ mV}$$

$$= \underline{442\text{ mV}}$$

Question 3.3 [4]

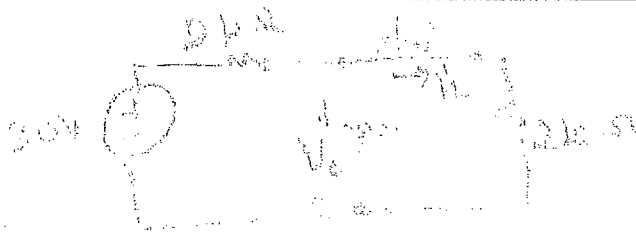
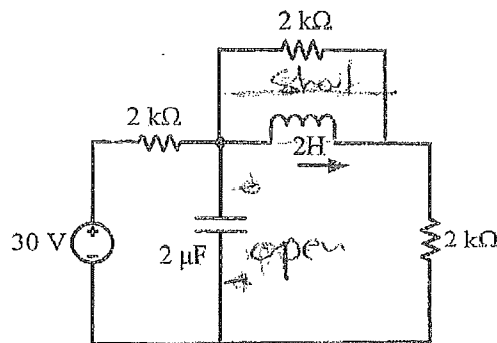
For the circuit below, assume DC conditions and calculate:

- The energy, w_C , associated with the capacitor.
- Current, i_L , flowing through the inductor.

Vraag 3.3 [4]

Vir onderstaande stroombaan, aanvaar gelykstrom toestand en bereken:

- Die energie w_C geassosieer met die kapasitor
- Die stroom, i_L , wat deur die inductor vloei.



$$\begin{aligned}
 a) \quad w_C &= \frac{1}{2} C v_C^2 \\
 &= \frac{1}{2} \times 2 \mu \times (15)^2 \\
 &= \underline{\underline{225 \mu J}}
 \end{aligned}$$

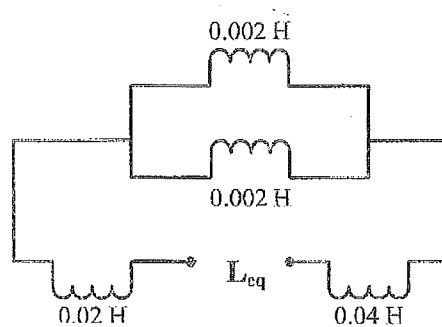
$$b) \quad i_L = \frac{30}{2k + 2k} = \frac{30}{4k} = \underline{\underline{7.5 \text{ mA}}}$$

Question 3.4 [2]

Calculate the equivalent inductance, L_{eq}

Vraag 3.4 [2]

Bereken die ekwivalente inductansie L_{eq} .



$$0.002 \parallel 0.002 \text{ (eq)} \\ = 0.001 \text{ H}$$

$$L_{eq} = 0.02 + 0.001 + 0.04 \\ = 0.061 \text{ H}$$

$$L_{eq} = 0.02 + 0.001 + 0.04 \\ = 0.061 \text{ H} \quad \checkmark \checkmark$$

Question 3.5 [2]

What is the voltage, $v_L(t)$, associated with an inductor of 2 mH, when the current through the inductor is $i(t) = (3.6 e^{-0.2t} + 2)$ A?

Vraag 3.5 [2]

Wat is die spanning $v_L(t)$ geassosieer met 'n induktor van 2mH indien die stroom deur die induktor $i(t) = (3.6e^{-0.2t} + 2)$ A is?

$$\begin{aligned} v_L(t) &= L \frac{d}{dt} i(t) = 2 \text{ m} \frac{d}{dt} (3.6 e^{-0.2t} + 2) \\ &= 72 \text{ m} \left[e^{-0.2t} \right] \times -0.2 \\ &= -1.44 e^{-0.2t} \text{ mV} \end{aligned}$$

Question 3.6 [2]

Simplify the complex number and present your answer in rectangular form.

$$Z_{\text{rect}} = \frac{2 + j3}{3 \angle 24^\circ} + 2 \angle 31^\circ + 2 - j3$$

Vraag 3.6 [2]

Vereenvoudig die komplekse getal en gee jou antwoord in reghoekige vorm.

$$Z_{\text{rect}} = \frac{2 + j3}{3 \angle 24^\circ} + 2 \angle 31^\circ + 2 - j3$$

$$\begin{aligned} &= \frac{3.6 \angle 56.3^\circ}{3 \angle 24^\circ} + (1.71 + j1.1) + 2 - j3 \\ &= 1.2 \angle 32.3^\circ + 3.71 - j1.97 \\ &= 1.01 + j0.64 + 3.71 - j1.97 \\ &= \underline{\underline{4.72 - j1.33}} \end{aligned}$$